

MIDTERM I

ECO403/503–INTRODUCTION TO CONTRACT THEORY AND AUCTIONS

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Instructions. The exam has a total of 120 points. I will grade it over 100, so there is the possibility to leave some items unanswered and still get the top grade of 100; alternatively, there is also the chance of obtaining 20 extra points above the top grade. The points of each question are shown in boldface. *You need to submit your written solutions in pdf format to Moodle. The submission also consists of a short video, strictly less than 15 minutes, in which you record yourself explaining to me your solutions. The deadline is Friday, October 30 at 11:59:59 PM. Good luck!*

1. **(40 points)** Consider the following family of preference relations defined over $L(Z)$ (the set of all lotteries with prizes in some finite set Z): The DM has in mind a function which assigns to each prize $z \in Z$ a value $v(z)$. He partitions Z into two sets G and B such that if $g \in G$ and $b \in B$ then $v(g) > v(b)$. He evaluates any lottery p by

$$p(\text{supp}(p) \cap G) \times \left(\max_{z \in \text{supp}(p) \cap G} \{v(z)\} \right) + p(\text{supp}(p) \cap B) \times \left(\min_{z \in \text{supp}(p) \cap B} \{v(z)\} \right).$$

These evaluations form his preferences over $L(Z)$, where $p(A) = \sum_{z \in A} p(z)$, i.e., given lottery p , the probability attached to prizes in the set A is $\sum_{z \in A} p(z)$.

- (a) **(10 points)** Explain the procedure in words.
(b) **(10 points)** Show that such a preference relation satisfies neither the Independence axiom nor the Continuity Axiom.
(c) **(10 points)** Show that a weaker independence property holds: if $\text{supp}(p) = \text{supp}(q)$ then for every $0 < \alpha < 1$ and every $r \in L(Z)$,

$$p \succsim q \Leftrightarrow \alpha p + (1 - \alpha)r \succsim \alpha q + (1 - \alpha)r$$

- (d) **(10 points)** Describe in words and then formally define a “monotonicity property” that holds.

2. **(40 points)** Consider a risk-averse individual who is an expected utility maximizer with a Bernoulli utility function over wealth $u(\cdot)$. The individual has initial wealth W and faces a probability θ of suffering a loss of size L , where $W > L > 0$. An insurance contract may be described by a pair (c_1, c_2) , where c_1 is the amount of wealth the individual has in the event of no loss and c_2 is the amount the individual has if a loss is suffered. That is, in the event no loss occurs the individual pays the insurance company an amount $(W - c_1)$, whereas if a loss occurs the individual receives a payment $[c_2 - (W - L)]$ from the company.¹

- (a) **(20 points)** Suppose that the individual’s only source of insurance is a risk-neutral monopolist, i.e., the monopolist seeks to maximize its expected profits. Characterize the contract the monopolist will offer the individual in the case in which the individual’s probability of loss, θ , is observable.

¹Hint: It might be helpful to draw a picture in (c_1, c_2) space. To do so, start by locating the individual’s endowment point, that is, what he gets if he does not purchase any insurance. When solving for the second-best, think about which agent would like to mimic the policy of the other agent to recognize which incentive constraints are binding and who is the “high” type as in the lectures. Also, note that the “outside option” of the individual if he doesn’t insure is not zero, but instead the utility obtained by not insuring.

- (b) **(20 points)** Suppose, instead, that θ is not observable by the insurance company (the individual knows θ). The parameter θ can take one of two values $\{\theta_L, \theta_H\}$, where $\theta_H > \theta_L > 0$ and $\Pr(\theta_L) = \lambda$. Characterize the optimal contract offers of the monopolist. Can one speak of one type of insured individual being “rationed” in his purchases of insurance (i.e., he would want to purchase more insurance if allowed to at fair odds)? Intuitively, why does this rationing occur?
3. **(40 points)** A government agency writes a procurement contract with a firm to deliver q units of a good. The firm has constant marginal cost c , so that its profit is $P - cq$, where P denotes the payment for the transaction. The firm’s cost is private information and the agency’s prior belief about c is distributed according to the pdf $f(c)$ on the interval $[\underline{c}, \bar{c}]$. The agency makes a take-it-or-leave-it offer to the firm (whose default profit is zero). Suppose that $B(q)$ denotes the benefit to the agency of obtaining q units. $B(q)$ is a strictly increasing and a strictly concave function of q .²
- (a) **(10 points)** Characterize the first-best contract.
- (b) **(10 points)** Characterize the second-best contract.
- (c) **(10 points)** Assume that $f(c)$ is the uniform distribution on $[\underline{c}, \bar{c}]$. With this specific distribution, characterize as far as you can parts (a) and (b).
- (d) **(10 points)** Assume that $f(c)$ is $\frac{2(\bar{c}-c)}{(\bar{c}-\underline{c})^2}$. With this specific distribution, characterize as far as you can parts (a) and (b).

²Hint: the more efficient firm in this problem would be the “high type” as in the lecture notes.